Day 8 – Camera Integration & Computer Vision Pipeline

1. Why is a camera required in a self-driving car?

A camera captures real-time images of the road and surroundings. These images help detect lane markings, traffic signs, traffic lights, pedestrians, vehicles, and obstacles so the vehicle can make safe driving decisions.

2. Why do we use ROI instead of the complete image?

ROI processes only the important road region where lane markings appear. This reduces computation, improves speed, and avoids processing irrelevant areas like the sky and buildings.

3. Which ROI percentage worked best and why?

The bottom 40% ROI worked best because it captured the lane markings while excluding most unnecessary parts of the image, providing a good balance between accuracy and efficiency.

4. Why is HSV preferred over RGB for color detection?

HSV separates color (Hue) from brightness (Value), making color detection more reliable under different lighting conditions. Yellow lane markings are easier to isolate using Hue thresholds.

5. What is image masking?

Image masking selects only the pixels that satisfy a specified condition, such as a color range, while suppressing all other pixels.

6. What is the purpose of Gaussian Blur?

Gaussian Blur reduces image noise and smooths the image before further processing, improving the quality of lane detection.

7. What is the difference between Morphological Opening and Closing?

Opening (erosion followed by dilation) removes small noise. Closing (dilation followed by erosion) fills small holes and connects broken lane markings.

8. Which preprocessing method produced the cleanest result?

The combination of Gaussian Blur, Morphological Opening, and Morphological Closing produced the cleanest mask by reducing noise and creating continuous lane markings.

9. Which OpenCV functions did you use?

cv2.cvtColor(), cv2.imshow(), cv2.waitKey(), cv2.destroyAllWindows(), cv2.inRange(), cv2.GaussianBlur(), and cv2.morphologyEx().

10. What did you learn from today's task?

I learned how to integrate a camera in Webots, capture live images, extract a Region of Interest, compare RGB, Grayscale, and HSV color spaces, detect yellow lane markings using HSV thresholding, and improve the results using Gaussian Blur and morphological operations.